

UEI501: CONTROL SYSTEMS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To understand concepts of the mathematical modelling, feedback control and stability analysis in Time and Frequency domains. The concept of time response and frequency response of the system will be studied.				
Syllabus Basic Concepts: Historical review, Definitions, Classification, Relative merits and demerits of open and closed loop systems, Linear and non-linear systems, Transfer function, Mathematical modelling of physical systems, Block diagrams and signal flow graphs. Components: D.C. and A.C. Servomotors, D.C. and A.C. Tach generators, Potentiometers and optical encoders, Synchro and stepper motors Analysis: Steady-state errors and error constants, Concepts and applications of P, PD, PI and PID types of control. Stability: Definition, Routh-Hurwitz criterion, Root locus techniques, Nyquist criterion, Bode plots, Relative stability, Gain margin and phase margins. Compensation: Lead, Lag and lag-lead compensators, Design of compensating networks for specified control system performance. State Space Analysis: Concepts of state, State variables and state models, State space equations, Transfer function, Transfer model, State space representation of dynamic systems, State transition matrix, Decomposition of transfer function, Controllability and observability.				
Laboratory Work: Linear system simulator, Compensation design, D.C. position control and speed control, Synchro characteristics, Servo demonstration, Stepper motor, Potentiometer error detector, Rate control system, Series control system, Temperature control system.				
Course Learning Objectives (CLO) The students will be able to: 1. Develop the transfer function of the physical systems. 2. Analyse the response of the closed and open loop systems. 3. Analyse the stability of the closed and open loop systems. 4. Design the various kinds of compensator. 5. Develop and analyse state space models.				
Text Books 1. Digital Control System, Gopal, M., Wiley Eastern (1986). 2. Control System Engineering, Nagrath, I.J. and Gopal, M., New Age International (P) Limited, Publishers (2003). 3. Modern Control Engineering, Ogata, K., Prentice Hall of India Private Limited (2001).				

Reference Books

1. Automatic Control System, Kuo, B.C., Prentice Hall of India Private Limited (2002).
2. Control System, Sinha, N.K., New Age International (P) Limited, Publishers (2002).

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30